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Photosensitization of nanocrystalline TiO_2 film electrode with cadmium sulphoselenide

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Abstract

Nanocrystalline titanium dioxide (TiO_2) thin films composed of densely packed nanometer-sized grains have been successfully deposited onto an indium-doped-tin oxide (ITO) substrate. Then cadmium sulphoselenide (CdSSe) thin film was deposited onto pre-deposited TiO_2 to form a TiO_2/CdSSe film, at low temperature using a simple and inexpensive chemical method. The X-ray diffraction, selected area electron diffraction, scanning electron microscopy (SEM), energy dispersive X-ray analysis (EDX) and water contact angle techniques were used for film characterization. Purely rutile phase of TiO_2 with super-hydrophilic and densely packed nanometer-sized spherical grains of approximate diameter 30–40 ($\pm$2) nm was observed. The increase in optical absorption was observed after CdSSe film deposition. Nest like surface morphology of CdSSe on TiO_2 surface results in air trapping in the crevices which prevents water from adhering to the film with increase in water contact angle. Photosensitization of TiO_2 with CdSSe was confirmed with light illumination intensity of 80 mW/cm^2.
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Keywords: TiO_2-CdSSe; Surface morphology; Optical absorption; *I–V*; Photoelectrochemical cells

1. Introduction

Nanocrystalline materials display unusual chemical and physical properties which are promising for the fabrication of novel nanodevices [1]. TiO_2 is one of the most extensively studied oxides because of its remarkable optical and electronic properties. Due to the strong confinement effects, the electrical and optical properties of these nanocrystalline thin films are varying significantly with the grain size. This ability reinforced to tune the physical properties of nanocrystalline thin films makes them an important category of materials that can find potential applications in the area of photoelectronic materials [2] with additional merits including: the low cost, more abundance in nature and various applications such as photocatalysis [3], photovoltaics [4], gas sensors [5] and more over in pharmaceuticals sciences as well [6–9]. An effective light interaction, thin TiO_2 films have been used as coating for self-cleaning glasses and tiles. Solution growth of nanocrystalline TiO_2 thin films from titanyl sulphate precursor solution using microwave heating chemical bath deposition is reported in the literature [10]. In the field of environmental protection, thin nanocrystalline TiO_2 films have many applications such as sterilization and sewage disposal. It shows good blood compatibility and can be used as artificial valves. In addition to organic dyes, narrow band gap semiconductor nanocrystals, such as CdS [11], CdSe [12,13], InP [14] and PbS [15], have attracted significant interest as light harvesting. The use of semiconductor nanocrystalline thin films as a sensitizer has some advantages in solar cell applications as (a) the band gap of the nanocrystals can be tuned by controlling their size to that the absorption spectrum match the spectral distribution of sunlight and a full-spectrum solar cell can be fabricated more easily, (b) nanocrystals have a large extinction coefficient due to their quantum confinements and intrinsic dipole moments, leading to rapid charge separation and (c) semiconductor nanocrystals have a robust inorganic nature [16]. In this paper, we wish to report synthesis and characterization of nanocrystalline, densely packed TiO_2 thin film on indium-doped-tin oxide (ITO) substrate at room temperature and formation of CdSSe thin layer for sensitization of pre-deposited TiO_2 with a conclusion that the photoelectrochemical (PEC) conversion efficiency and stability of bilayer films are higher than individuals.

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2. Experimental details

2.1. Deposition of nanocrystalline TiO_2 and TiO_2/CdSSe thin films

The method of preparation of TiO_2 nanocrystalline thin films is recently reported [17]. Titanium (III) chloride ($TiCl_3$) liquid was mixed with water in appropriate quantities. Specifically, 2 mL of $TiCl_3$ (20–30 wt% in HCl, Aldrich, USA) was added to an ice-cooled of 20 mL de-ionized water (18 MΩ cm). The pH was adjusted between 3.0 to 3.5 using 7% ammonium hydroxide solution with a constant stirring of precursive solution. After stirring for 30 min, a homogeneous pale violet-blackish solution was obtained, which can be stable for several hours at room temperature, i.e. 300 K. Change in color of the resulting homogeneous solution depends on the pH as a result of deprotonation by water molecules of the peroxotitanium complex which is reported elsewhere [18]. The ITO substrates were carefully rinsed with de-ionized water, ultrasonic cleaner and dried with the argon gas before use. Solution color changed from violet-blackish to whitish after 6 h and further dipping of ITO substrate in the bath resulted in dissolution of both TiO_2 and ITO coatings from the substrate in the solution. For the deposition of CdSSe films onto previously deposited TiO_2 films, 0.1 M solutions each of cadmium chloride, thiourea and sodium selenosulphate were used together (complexed with ammonia, pH 10 (±0.2). In the presence of Cd^{2+} ions in the bath, CdSSe was formed as the ionic product of Cd^{2+}, S^{2-} and Se^{2-} exceeds the solubility product of CdSSe. In the preset study, the depositing conditions were same as reported earlier [19] (i.e. deposition temperature 333 K for 45 min).

2.2. Characterization techniques

Phase present in as-deposited TiO_2 and TiO_2/CdSSe thin films, chemical elemental compositions, surface morphologies were examined using X-ray diffraction (XRD) (Cu Kα, 40 kV, 100 mA), selected area electron diffraction (SAED), scanning electron microscopy (SEM) (operated at 15 kV), energy dispersive X-ray analysis (EDX), optical absorption, respectively. The thickness of the TiO_2 and TiO_2/CdSSe films was measured by laser ellipsometry with an incident angle of 70° and wavelength 632.8 nm after the single crystal Si standardization. Optical absorption spectra were recorded at room temperature using a 100 CONC spectrophotometer in the wavelength range of 350–850 nm. The wettability on TiO_2 and TiO_2/CdSSe surfaces were evaluated by water contact angle measurements, using a commercial contact angle meter (Surface Electo-Otpics, Model-Phoenix 150). Low intensity lamp (2 mW cm^{-2}) was used for measurement. The 2 mL drops were formed at different positions on the film surface using a Rame-Hart Inc. model-10 micro-syringe. Contact angle was determined 30 s after the formation of water drop on the film. The photo-electrochemical setup consisted of a quartz cell equipped with a working-electrode (TiO_2/CdSSe) and a Pt counter electrode. 1-hexyl-2-3-dimethyl-imidazolium iodide (C6DMI, 0.6 M), LiI (0.1 M), I_2 ((0.05 M) and 4-tbutylpyridine (*t*-Bpy, 0.5 M) were dissolved in 15 mL methoxyacetonitrile to

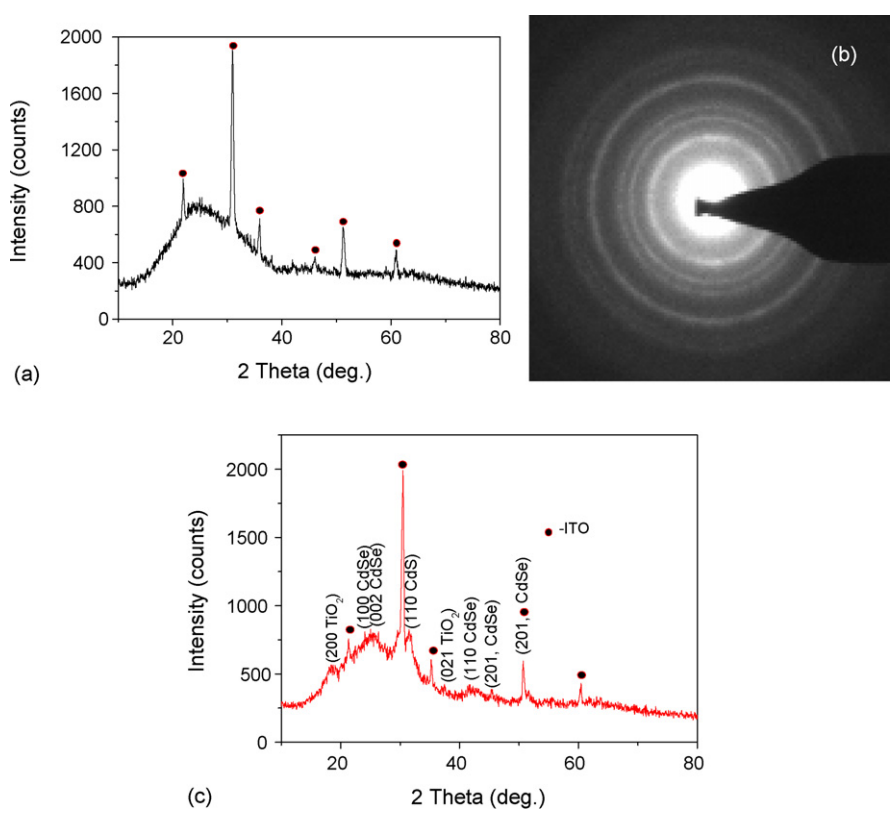

Fig. 1. The XRD (a) and SAED (b) patterns of TiO_2 nanocrystalline film and the XRD of pattern of TiO_2 after CdSSe film formation (c).

obtain an organic electrolyte solution for the photoelectrochemical cell. The current voltage (*I–V*) measurement was carried out using 1 kW xenon lamp with photo-intensity of 80 mW/cm^2. The electrode size was 0.28 cm^2, unless otherwise noted.

3. Results and discussion

3.1. Structural characterization, surface morphology and optical absorption

As-deposited TiO_2 film was violet in color. It is well-known that the titanium in aqueous solution can form a series of colored peroxotitanium complexes depending on the pH of the solution [18]. At low pH, it is generally accepted that the ratio of peroxo (O–O) groups to titanium in the species is 1:1, although the exact species in the acid solutions remain controversial [20]. In the present case, as pH of solution was low, we believe, hydrolyzed titanium cations would lead to hydrous oxide precipitation by condensing through the olation and oxolation as reported by Gao et al. [21], which may be the one of the possible causes for strong absorption of light in visible spectrum (optical absorption section). The CdS [22] and CdSe [23,24] films are known for their yellow and dark-pink colors, respectively. After mixing the solutions together in 1:1 ratio, brown colored film was observed, confirming the formation of cadmium sulphoselenide as also supported by XRD and EDX. The film thicknesses of TiO_2 and CdSSe over TiO_2 were 150 (±5) nm and 300 (±5) nm, respectively.

The deposition of CdSSe on theTiO_2 was partially supported by its brown color, confirmed by XRD and EDX analyses. The XRD pattern of ITO/TiO_2 [Fig. 1(a)] is not of much significant for the structural study as background peaks of ITO are more dominant which makes difficult to calculate grain size using the Scherrer′s formula. For confirming the type of TiO_2 and its phase, SAED pattern [Fig. 1(b)] is presented. The existence of well-defined sharp rings is good support for nanocrystalline nature whereas for amorphous films, fuzzy and broad rings generally observed [25]. The SAED pattern shows a set of rings instead of spots due to random orientation of crystallites, corresponding to the diffraction from different planes of nanocrystallites. The location of planes corresponding to such as (1 1 0), (1 1 2), (1 1 1), (2 1 1), (0 5 2) and (3 0 1) are in good agreement with the Joint Committee on Powder Standard Diffraction (JCPDS) [26] reference diagram for the corresponding bulk phases with lattice constants $a = 4.593$ Å, and $c = 2.959$ Å. The rings are corresponding to specific lattice planes of the nanocrystallites. The peak positions index well with the bulk rutile TiO_2 lattice reflections. Diffraction peaks [Fig. 1(c)] at $2\theta = 31.79$ and 23.90, 25.35, 41.965, 45.784 and 50.671° correspond to the (1 1 0) for CdS [27], and (1 0 0), (0 0 2), (1 1 0), (1 0 3) and (2 0 1) panes for CdSe [28], respectively. This observation confirmed the formation of mixed phase as CdSSe.

Fig. 2(a and b) shows the SEM images of as-deposited TiO_2 and TiO_2/CdSSe films. It is seen that the TiO_2 film surface is smooth, homogeneous and well-covered consisting densely packed spherical grains of 30–40 (±2) nm diameter without the formation of any voids, pinholes and cracks. After CdSSe film deposition, film surface morphology has been considerably changed to nest-like. This morphological change is attributed to the presence of excess cadmium in atomic percentage [29]. The measurement of EDX carried out for an elemental chemical stoichiometric analysis involved in this films was limited to a selected elements such as Cd, S and Se as other elemetals Ti, O, In, Sn were not of present study interest [Fig. 2 (inset)]. Chemical analysis has been carried out only for Cd, S and Se elements as they are of our interest. The average atomic percentage was found to be 58:23:19, showing that the film was in good stoichiometric ratio with selenium excess. This result has been attributed to the fact that the solubility product of CdS is smaller than that of CdSe [30].

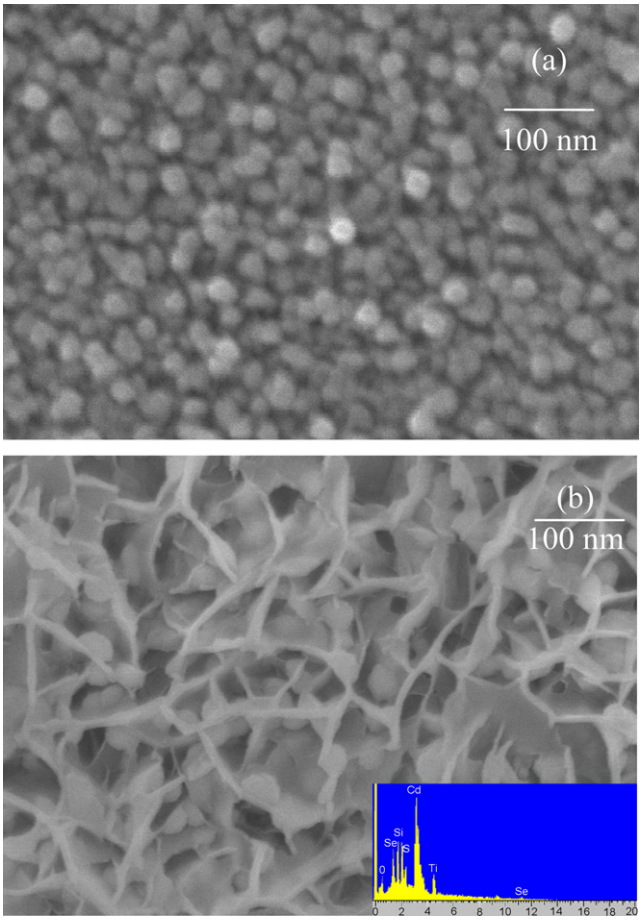

Fig. 2. The SEM images of the as-deposited TiO_2 (a) and that of TiO_2/CdSSe (b). Inset of (b) shows typical EDX pattern recorded while SEM measurement.

The optical absorption spectra shown in Fig. 3 elucidate that the as-deposited TiO_2 film was highly transparent with excitonic peak (due to size quantization and/or transition from the spin-orbit split valence band to the conduction band) in the visible region. After CdSSe deposition, drastic and considerable absorption increase in visible region was noticed.

3.2. Surface wettability

The wetting of solid with water, where air is the surrounding medium, is dependent on the relation between the interfacial

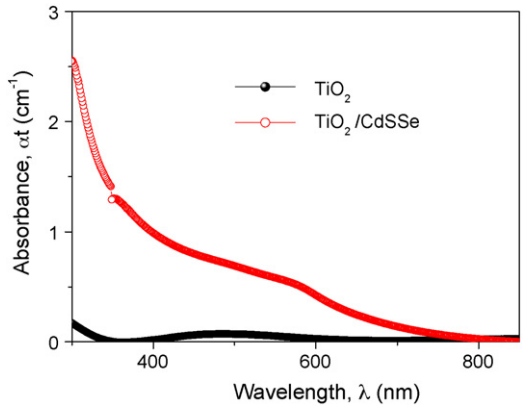

Fig. 3. The variation in optical absorbance for TiO_2 and TiO_2/CdSSe film.

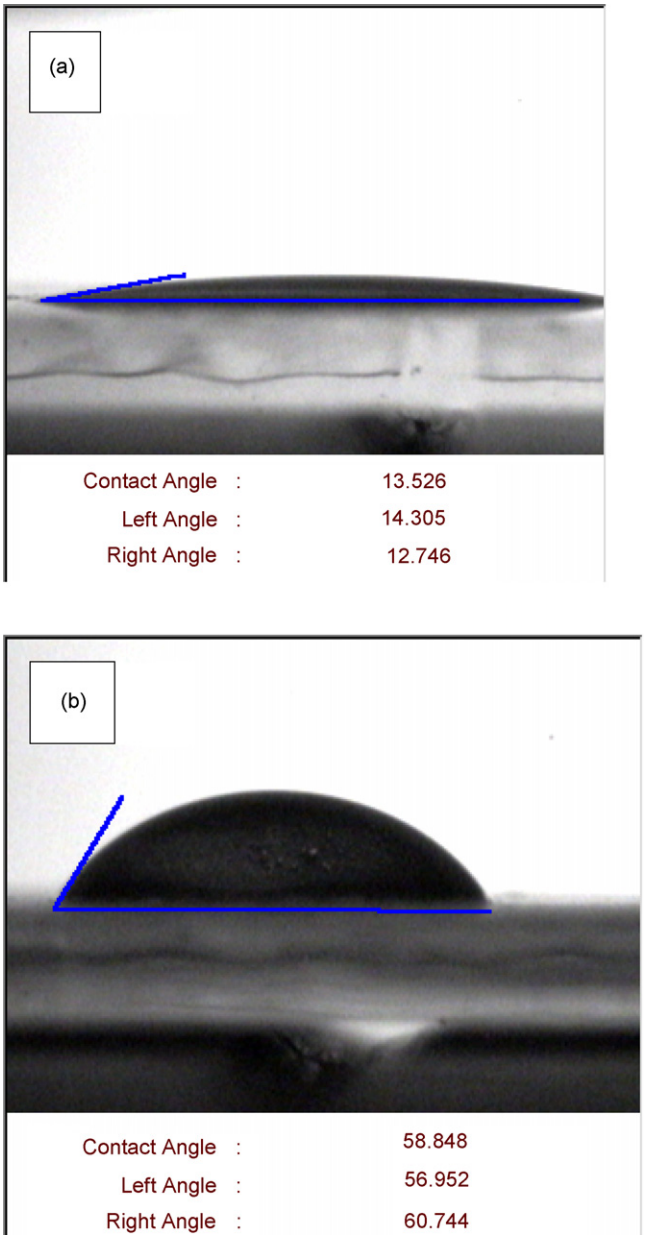

Fig. 4. Measurement of surface water contact angles on TiO_2 (a) and TiO_2/CdSSe (b) films.

tensions (water/air, water/solid and solid/air). The ratio between these tensions determines the contact angle (θ) between a water droplet on a given surface. A contact angle of 0° means complete wetting and a contact angle of 180° corresponds to complete non-wetting. Both super-hydrophilic and super-hydrophobic surfaces are important for practical applications [31]. In the present case, the water lies flat with contact angle of 13 (±1.5)° on the surface of TiO_2 film in sheets instead of forming droplet as seen in Fig. 4 (a). This specific property is attributed to nanocrystalline nature that is expected to possess very high surface energy, which increases with reduction in particle size. Direct relationship of surface energy with water surface tension is well-known. High wettability gives rise to small water contact angle which has direct surface tension relation. It is reported that nanocrystalline thin films show high surface energy due to the presence of uniform and non-uniform strains [32]. After CdSSe film deposition, the water contact angle was increased to 58 (±1.5)° [Fig. 4(b)]. Fibrous and/or nest like surface morphology of CdSSe might have caused for higher water contact angle value [33].

3.3. Photo-electrochemical cell (PEC) studies

The *I–V* characteristics of TiO_2/CdSSe film electrode were obtained in dark and upon illumination condition. Fig. 5 shows photocurrent versus applied potential curves of TiO_2/CdSSe film electrode in lithium iodide electrolyte. Compared with the bare TiO_2 (not shown) film electrode, the TiO_2/CdSSe system exhibited the increased performance. Upon optical excitation of CdSSe, the photo-generated electrons are quickly transferred to TiO_2 while the holes accumulate at the CdSSe grains. If the holes quickly reach the surface and react with the species in the electrolyte, the excited electrons left behind will diffuse to the back contact where they are withdrawn as a photocurrent. It means that the electrons excited in the conduction band of CdSSe are injected to the conduction band of TiO_2. In the coupled system, the conduction band of TiO_2 practically acts as an electron acceptor that makes the electron transfer quickly to TiO_2

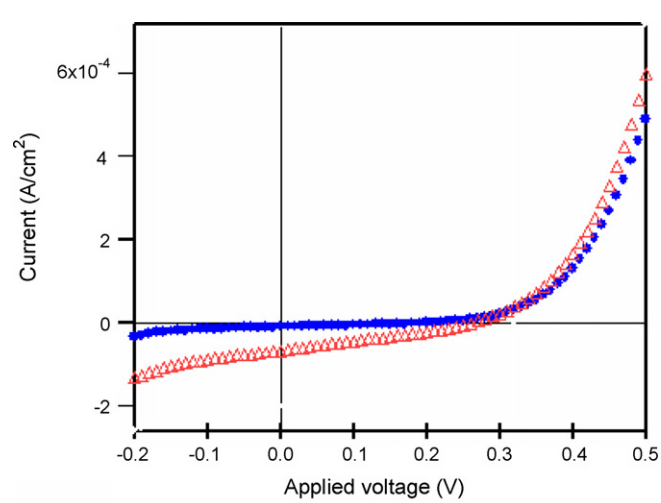

Fig. 5. The *I–V* dark (dots) and light (triangle) curves of TiO_2/CdSSe film electrode.

conduction band [34]. Here, we believe two possible roles of TiO_2 as: (a) it acts as a back contact due to its high band gap energy (hole blocker), which avoids electron-hole recombination at ITO and (b) provides its conduction band for the charge transport which avoids photo-corrosion. Under illumination at the junction the magnitude of voltage increases with negative polarity towards TiO_2/CdSSe electrode and cathodic behavior observed and approaches to saturation as is commonly observed in n-type semiconductor electrodes. The observed low conversion efficiency (0.035%) is attributed to the difference in grain structure at TiO_2 and CdSSe interface (large recombination losses). The photo-generated electrons in semiconductor film either recombine readily with the holes or leak out at the electrode interface; instead of flowing through the external circuit, deep electron trapping in the CdSSe [35] (or oxide substrate electrode) and potential dependent electron injection rates [36] may be the possible reasons for the low conversion efficiency.

4. Conclusions

In this paper, we have illustrated the photosensitization of compact, densely packed, nanocrystalline TiO_2 electrode by ternary semiconductor CdSSe film. Violet colored TiO_2 film becomes brown after CdSSe deposition, whose formation takes place via chemical deposition. An enhancement in water contact angle after CdSSe deposition is attributed to the nest-like surface morphology. It is concluded that the CdSSe-sensitized TiO_2 nanostructured electrode shows photoactivity in PEC cells.

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